UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 12

Content.

1. Sub-Exponential Distributions Continued
2. Maximal Inequalities

1 Sub-Exponential Distributions Continued

Recall that X is sub-exponential with (σ^2, α) if for $\mu = \mathbb{E}X$,

$$\mathbb{E}\left[e^{s(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2 s^2}{2}\right) \quad \text{for } |s| < \frac{1}{\alpha}.$$

Theorem 1. Suppose $X_1, \cdots, X_N$ are independent, centered ($\mathbb{E}X_i = 0, \forall i$) sub-exponential with $(1, 1)$. Then, for $a = (a_1, \cdots, a_N) \in \mathbb{R}^N$, we have

$$\sum_{i=1}^{N} a_i X_i \sim \text{subExp}(\|a\|_2^2, \|a\|_\infty).$$

Therefore,

$$\mathbb{P}\left[\left|\sum_{i=1}^{N} a_i X_i\right| > t\right] \leq 2\exp\left(-\min\left\{\frac{t^2}{2\|a\|_2^2}, \frac{t}{2\|a\|_\infty}\right\}\right).$$

Example : Let $a_i = \frac{1}{N}$ for all i, then

$$\mathbb{P}\left[\left|\frac{1}{N}\sum_{i=1}^{N} X_i\right| > t\right] \leq \exp\left(-N\min\left\{\frac{t^2}{2}, \frac{t}{2}\right\}\right),$$

$$\mathbb{P}\left[\left|\frac{1}{N} X_i\right| > t\right] \leq \exp\left(-\frac{Nt}{2}\right).$$

So sub-exponential tails follow from "one bad event".

Example :

$$\mathbb{P}\left[\left|\frac{1}{\sqrt{N}}\sum X_i\right| > t\right] \leq \begin{cases} 2e^{-\frac{t^2}{2}} & t \leq \sqrt{N}, \\ 2e^{-\frac{\sqrt{N}t}{2}} & t > \sqrt{N}. \end{cases}$$

This means that for $t < \sqrt{N}$ (small regime), we have Gaussian tails; for large t, heavier tails kick in.

Theorem 2 (Sub-Exponential Properties)**.** Let X be a random variable. Then the following are equivalent, where the parameters differ by at most absolute constants.

1. (Tails)
$$\mathbb{P}\left[|X| \geq t\right] \leq 2e^{-\frac{t}{k_1}}, \quad \forall t \geq 0.$$
2. (Moments)
$$\|X\|_{L^p} = (\mathbb{E}|X|^p)^{1/p} \leq k_2 p, \quad \forall p \geq 1.$$
3. (MGF of $|X|$)
$$\mathbb{E}e^{\lambda|X|} \leq e^{k_3\lambda}, \quad \forall \lambda \in \left[0, \frac{1}{k_3}\right].$$
4. (Bound on MGF)
$$\mathbb{E}e^{\frac{|X|}{k_4}} \leq 2.$$

If, moreover, $\mathbb{E}X = 0$, then the above are equivalent to

5. (MGF of X)
$$\mathbb{E}e^{\lambda X} \leq \exp\left(k_5^2\lambda^2\right), \quad \forall \lambda \leq \frac{1}{k_5}.$$

Definition 1 (Orlicz Norm)**.**

$$\|X\|_{\psi_1} := \inf\left\{t > 0 : \mathbb{E}\exp\left(\frac{|X|}{t}\right) \leq 2\right\}.$$

Lemma 1. Let X, Y be sub-Gaussian. Then

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2}\|Y\|_{\psi_2}.$$

Proof. We assume $\|X\|_{\psi_2} = \|Y\|_{\psi_2} = 1$. Recall that

$$\|X\|_{\psi_2} = \inf\left\{t > 0 : \mathbb{E}\exp\left(\frac{X^2}{t^2}\right) \leq 2\right\}.$$

We have

$$\begin{aligned}
\mathbb{E}\left[e^{|XY|}\right] &\leq \mathbb{E}\left[e^{\frac{X^2}{2}+\frac{Y^2}{2}}\right] \quad \text{(by AM–GM)} \\
&\leq \mathbb{E}\left[\frac{1}{2}e^{X^2} + \frac{1}{2}e^{Y^2}\right] \quad \text{(by AM–GM)} \\
&= \frac{1}{2}\mathbb{E}e^{X^2} + \frac{1}{2}\mathbb{E}e^{Y^2} \leq 1 + 1 = 2.
\end{aligned}$$

So, $\|XY\|_{\psi_1} \leq 1$. □

Fact 1. $\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2$.

2 Maximal Inequalities

Theorem 3. Let $X_1, \cdots, X_N$ be sub-Gaussian with parameter σ^2. Define

$$X := (X_1, \cdots, X_N) \in \mathbb{R}^N. \quad \text{Then:}$$

(a) $\mathbb{E}\|X\|_2 \leq \sigma\sqrt{N}$,

(b) $\mathbb{E}\left[\max_i X_i\right] \leq \sigma\sqrt{2\log N}$,

(c) $\mathbb{E}\left[\max_i |X_i|\right] \leq \sigma\sqrt{2\log 2N}$,

(d) $\mathbb{P}\left[\|X\|_\infty > t\right] \leq 2N\exp\left(-\frac{t^2}{2\sigma^2}\right)$.

Proof. (a)

$$\begin{aligned}
\mathbb{E}\|X\|_2 &= \mathbb{E}\sqrt{\|X\|_2^2} \\
&\leq \sqrt{\mathbb{E}\|X\|_2^2} \quad \text{(Jensen)} \\
&= \sqrt{\sum_{i=1}^N \mathbb{E}X_i^2} \\
&\leq \sqrt{N \cdot \delta^2} \quad (\mathbb{E}X_i^2 \leq \sigma^2) \\
&= \delta\sqrt{N}.
\end{aligned}$$

(b)

$$\begin{aligned}
\mathbb{E}\left[\max_i X_i\right] &= \frac{1}{s}\mathbb{E}\left[\log e^{s\max_i X_i}\right] \quad \text{(for } s>0\text{)} \\
&\le \frac{1}{s}\log \mathbb{E}\left[e^{s\max_i X_i}\right] \quad \text{(Jensen)} \\
&\le \frac{1}{s}\log \mathbb{E}\left[\sum_{i=1}^N e^{sX_i}\right] \\
&= \frac{1}{s}\log\left(\sum_{i=1}^N \mathbb{E}e^{sX_i}\right) \\
&\le \frac{1}{s}\log\left(Ne^{\frac{s^2\sigma^2}{2}}\right) \quad (X_i\text{'s are sub-Gaussian}) \\
&= \frac{\log N}{s} + \frac{s\sigma^2}{2}.
\end{aligned}$$

Now we optimize over $s>0$. Set $s = \frac{\sqrt{2\log N}}{\sigma}$, then

$$\mathbb{E}\left[\max_i X_i\right] \le \frac{\delta\sqrt{2\log N}}{2} + \frac{\delta\sqrt{2\log N}}{2} = \delta\sqrt{2\log N}.$$

(c) Define $\overline{X}_1 = X_1, \ldots, \overline{X}_N = X_N, \overline{X}_{N+1} = -X_1, \ldots, \overline{X}_{2N} = -X_N$, so that

$$\max_i |X_i| = \max_{1\le j\le 2N} \overline{X}_j.$$

Applying (b) to $\overline{X}_j$ gives the result.

(d)

$$\begin{aligned}
\mathbb{P}\left[\|X\|_\infty > t\right] &= \mathbb{P}\left[\max_i |X_i| > t\right] \\
&\le \sum_{i=1}^N \mathbb{P}[|X_i| > t] \quad \text{(union bound)} \\
&\le 2N\exp\left(-\frac{t^2}{2\sigma^2}\right).
\end{aligned}$$

□

Remark. The theorem does not assume independence.

Remark. How about lower bounds? Is it necessarily true that

$$\mathbb{E}\left[\max_i |X_i|\right] = \Omega(\sqrt{\log N})?$$

No, e.g., if $X_1 = X_2 = \cdots = X_N$.

Fact 2. If $X_1, \ldots, X_N$ are i.i.d. standard Gaussian, then

$$\mathbb{E}\max_i |X_i| \sim \sqrt{2\log N}(1 + o_N(1)).$$

How about max over infinite sets?

Definition 2 (Polytope)**.** A polytope P is a compact set with finitely many vertices $V(P)$ such that

$$P = \mathrm{conv}(V(P)),$$

the convex hull of $V(P)$.

Let $X \in \mathbb{R}^d$ be a random vector and P be a polytope. We want to find

$$\sup_{\theta \in P} \theta^T X.$$

Lemma 2. Fix $c \in \mathbb{R}^d$, consider the linear map $x \mapsto c^T x$. Then,

$$\max_{x \in P} c^T x = \max_{x \in V(P)} c^T x,$$

where P is a polytope.

Proof. Clearly, $\max_{x \in V(P)} c^T x \le \max_{x \in P} c^T x$. Now fix $x \in P$. Let $V(P) = \{v_1, \ldots, v_N\}$, and let $\lambda_i \ge 0$ with $\sum_{i=1}^N \lambda_i = 1$ such that $x = \sum_{i=1}^N \lambda_i v_i$. Then

$$\begin{aligned} c^T x &= \sum_{i=1}^N \lambda_i c^T v_i \\ &\le \sum_{i=1}^N \lambda_i \max_{v \in V(P)} c^T v \\ &= \max_{v \in V(P)} c^T v. \end{aligned}$$

Hence, $\max_{x \in P} c^T x \le \max_{v \in V(P)} c^T v$. □

Note that if X has independent sub-Gaussian entries, then $c^T X$ is also sub-Gaussian.

Theorem 4. Let P be a convex polytope with N vertices $v^{(1)}, \ldots, v^{(N)}$, and $X \in \mathbb{R}^d$ be a random vector such that $(v^{(i)})^T X$ is sub-Gaussian with parameter σ_i^2 for $1 \leq i \leq N$. Then

$$\mathbb{E}\left[\max_{\theta \in P} \theta^T X\right] \leq \sigma\sqrt{2 \log N},$$

$$\mathbb{E}\left[\max_{\theta \in P} |\theta^T X|\right] \leq \sigma\sqrt{2 \log 2N}.$$

Definition 3 (ϵ-net)**.** Fix $K \subset \mathbb{R}^d$. A set $\mathcal{N} \subset K$ is called an ϵ-net for K if $\forall x \in K, \exists x' \in \mathcal{N}$ such that

$$\|x - x'\|_2 \leq \epsilon.$$